北京一零一中
BEIJING 101 HIGH SCHOOL

基于火灾危险气体释放与风场驱动二维对流-扩散模型的校园室外疏散方案设计方法：

以一零一中学为例

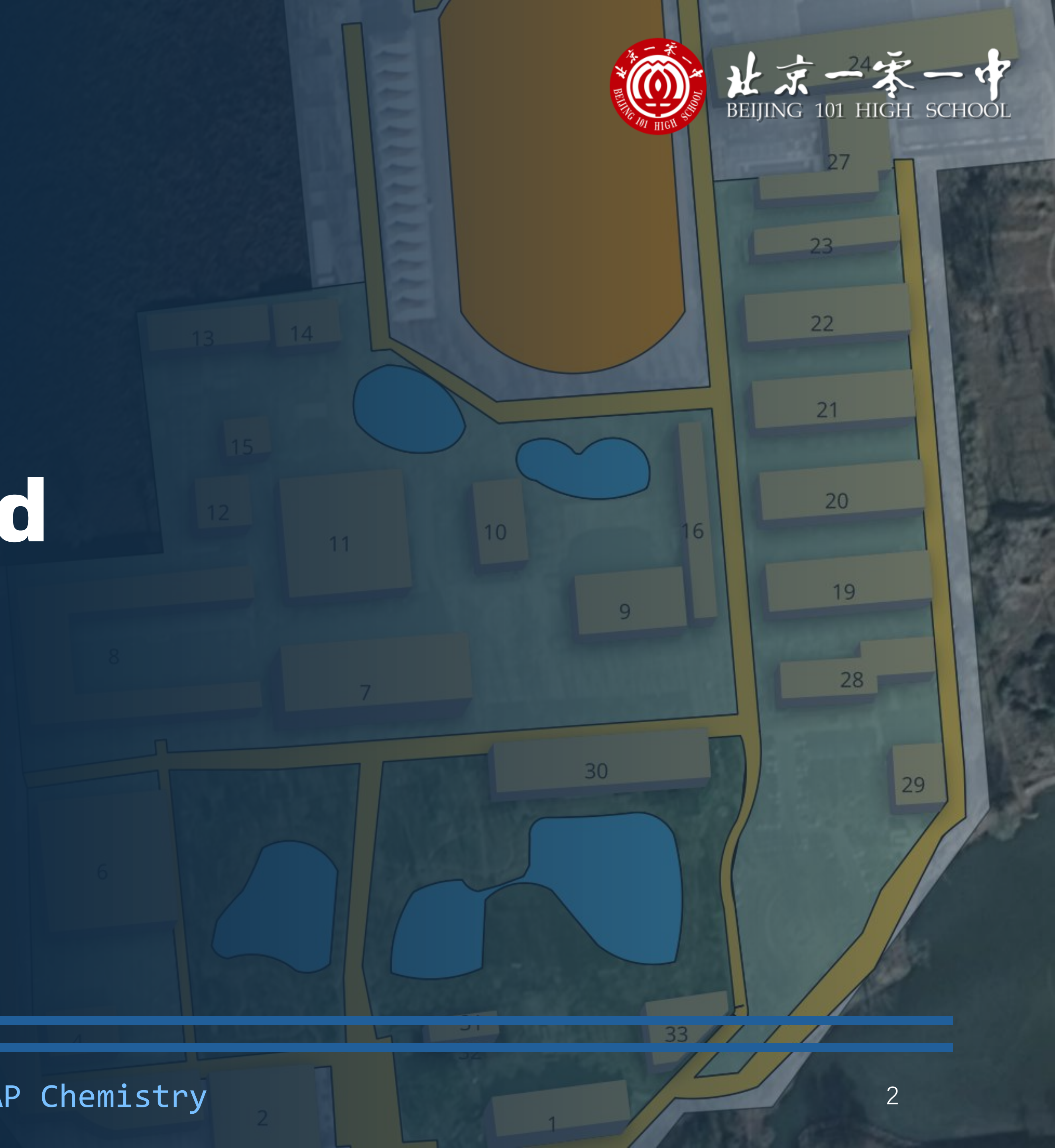
Design Method of Campus Outdoor Fire Evacuation Plan Based on Hazardous Gas Release and Wind-Driven 2D Advection-Diffusion Model:

A Case Study of Beijing 101 Middle School

开题报告
Research Proposal

Authors:
陈禹翔 Lambert Yuxiang Chen[1]
彭博 Blake Bo Peng[1]
[1]Beijing 101 Middle School International Department

Chapter 1.
Research Background
研究背景



背景与问题 Background & Question

校园火灾是一类具有突发性和高风险性的安全事件。对于学校这类人员密集场所而言，火灾发生后，如何在有限时间内将学生和教职工疏散到安全集合点，是应急管理中的重要问题。传统校园疏散方案通常更关注建筑内部逃生路线，例如楼梯、走廊和出口的选择；然而，在人员离开建筑后，校区内部的**室外疏散过程**同样会影响最终安全性。

Campus fires are sudden and high-risk safety events. In a school, where people are densely concentrated, it is crucial to evacuate students and staff to safe assembly points within a limited time after a fire breaks out. Traditional evacuation plans often focus on indoor escape routes such as stairways, corridors, and exits; however, the **outdoor evacuation process** after leaving the building can also affect the final safety outcome.

Fig 1. Split smoke plume
(Daniel Case, 2018)

研究问题 Research Problem

本课题关注的问题是：

在火灾危险气体释放和风场影响下，怎样选择更安全的校园室外疏散路径与集合点？

The problem that this project will focus on is:

Under the influence of fire hazard gas release and wind field, how to choose safer outdoor evacuation routes and assembly points on campus?

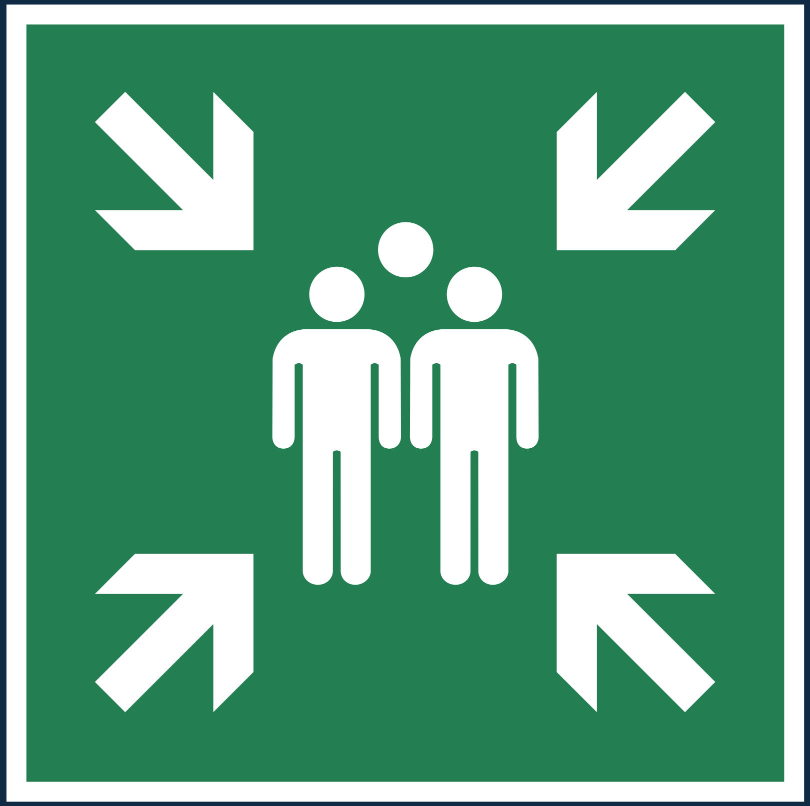


Fig 2. Assembly point icon by ISO 7010 E007 (MaxxL, 2026)

模型假设 Model Assumptions

为了控制研究规模，对模型做**适当假设**：

1. 本研究只讨论人员离开建筑后的校园室外疏散过程，着火建筑被简化为危险气体释放源，释放强度由燃烧材料类型、燃烧速率和气体生成系数估算；
2. 短时间疏散过程中，室外气体影响主要集中在着火建筑附近、出口周边、下风向道路和距离火源较近的集合点；
3. 实际模拟采用有限差分法，将物理场以网格表示及计算；
4. 校园道路、建筑出口和集合点将被抽象为图结构，路段容量和集合点容量在模型中视为已知或可估计参数；
5. 人员疏散行为按人群单元处理，暂不细分个体复杂行为。

To control the scale of the study, **appropriate assumptions** are made for the model:

1. This study only discusses the outdoor evacuation process on campus after personnel have left the building. The building on fire is simplified as a source of dangerous gas release, with the release intensity estimated based on the type of burning material, burning rate, and gas generation coefficient;
2. During a short-term evacuation process, the impact of outdoor gases is primarily concentrated near the burning building, around exits, on downwind roads, and at assembly points located closer to the fire source;
3. The actual simulation employs the finite difference method to represent and calculate physical fields using grids;
4. Campus roads, building exits, and assembly points will be abstracted into a graph structure, with road segment capacity and assembly point capacity treated as known or estimable parameters in the model;
5. Evacuation behavior is handled based on population units, without further breakdown into individual complex behaviors for the time being.

BEIJING 101 HIGH SCHOOL

Chapter 2.
Model Establishment and Solution
模型建立及求解

方案概述 Overview

BEIJING 101 HIGH SCHOOL

Step 1

建立燃烧危险气体产生模型。

根据AP Chemistry知识分析建筑材料燃烧时释放气体的成分及释放气体的量随着时间的变化关系。

Establish a model for the production of hazardous combustion gases.

Analyze the composition of gases released during the combustion of building materials and the relationship between the amount of gas released vs. time, based on AP Chemistry knowledge.

Step 2

建立室外危险气体扩散模型。

应用AP Calculus BC + CS知识，参考基于菲克定律的二维对流-扩散模型，在考虑风场影响的情况下计算气体扩散情况。

Establish an outdoor hazardous gas diffusion model.

Applying the knowledge of AP Calculus BC + CS, and referencing the two-dimensional convection-diffusion model based on Fick's law, the gas diffusion situation is calculated while considering the influence of the wind field.

Step 3

建立路径优化与目标函数模型。

整合Step2中的危险气体扩散模型及学校抽象图，综合构建 *Cost* 函数，并使用Computer Science算法求解以得到优化方案。

Establish a path optimization and objective function model.

Integrate the hazardous gas diffusion model from Step 2 with the abstract diagram of the school, comprehensively construct a Cost function, and use the Computer Science algorithm to solve for an optimized solution.

Step 1: 燃烧危险气体产生模型 The Model of the Production of Hazardous Gases by Combustion

北京一零一中 BEIJING 101 HIGH SCHOOL

1a. 成分分析 Analysis of Material Composition

建筑中的主要可燃成分为各种泡沫与塑料，还有堆放的纸张等。这些物质的化学成分大多包含C,H,O。以下列出了建筑材料燃烧的反应概括简式。

The main combustible components in buildings are various foams and plastics, as well as stacked paper, etc. The chemical composition of these substances mostly contains C, H, O. The following is a simplified summary of the combustion reactions of building materials.

When Oxygen is enough: $\quad \mathit{Combustible\ materials} + O_2(g) \rightarrow CO_2(g) + H_2O(g)$

When Oxygen is NOT enough: $\quad \mathit{Combustible\ materials} + O_2(g) \rightarrow \boldsymbol{CO(g) + smoke\ particles + toxic\ gases}$

For N-containing materials: N-containing $\mathit{Materials} + O_2(g) \rightarrow \boldsymbol{HCN(g)} + \boldsymbol{CO(g)} + H_2O(g)$

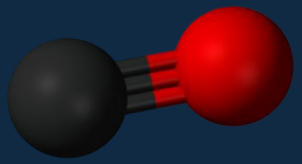


Fig 3. CO molecule 3-D model (Benjah-bmm27, 2007)

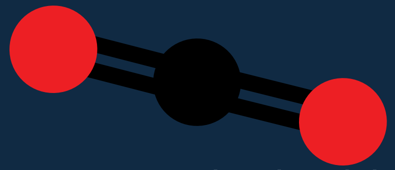


Fig 4. CO_2 molecule odel (Adenosine, 2013)

Step 1: 燃烧危险气体产生模型 The Model of the Production of Hazardous Gases by Combustion

BEIJING 101 HIGH SCHOOL 北京一零一中

1b. 简化气体生成模型 Simplified Gas Generation Model

将**气体释放速率 q_i** 近似为常数。因此，第 i 种**危险气体的累计生成质量**可近似为关于时间的一次函数。其中，q_i为第i种危险气体的平均释放速率，$m_i(t)$为时间t累计生成的气体质量。

Approximates the **gas release rate q_i** as constant. Therefore, **the cumulative mass of hazardous gas** i can be approximated as a linear function of time, where q_i is the average release rate of hazardous gas i, and $m_i(t)$ is the cumulative mass generated by time t.

$$m_i(t) \approx q_i t$$

1c. 气体源项计算 Gas source term calculation

扩散方程中的源项表示**单位时间内源区浓度的增加速率**，因此：

In the diffusion equation, the source term represents the rate of increase of concentration per unit time. Thus:

$$S_i(t) = \frac{\partial \frac{m_i(t)}{V_s}}{\partial t} = \frac{q_i}{V_s}$$

Step 2: 基于菲克定律的二维对流-扩散模型 Two-dimensional Convection-Diffusion Model Based on Fick's Law

北京一零一中 BEIJING 101 HIGH SCHOOL

室外危险气体扩散用**基于菲克定律的二维对流-扩散方程**描述。模型引入风速向量、有效扩散系数和衰减项，重点分析着火建筑附近与下风向区域的浓度变化。

Outdoor hazardous gas dispersion is described by a **two-dimensional advection-diffusion equation based on Fick's law**. The model introduces a wind vector, an effective diffusion coefficient, and a decay term. It focuses on concentration changes near the fire source and in downwind regions.

Fick's Second Law of Diffusion (Basic Expression):

Concentration

Diffusion Coefficient

$$\frac{\partial c}{\partial t} = D\frac{\partial^2 c}{\partial x^2}$$

Two-dimensional advection-diffusion equation of gas i:

$$\frac{\partial C_i}{\partial t} = \mathrm{D}_{\mathrm{eff},i}\left(\frac{\partial^2 C_i}{\partial x^2} + \frac{\partial^2 C_i}{\partial y^2}\right) - u\frac{\partial C_i}{\partial x} - v\frac{\partial C_i}{\partial y} + S_i(t) - \lambda_i C_i\,, \quad C_{0,i} = 0$$

Effective Diffusion Coefficient

(u, v) is the wind vector in the field

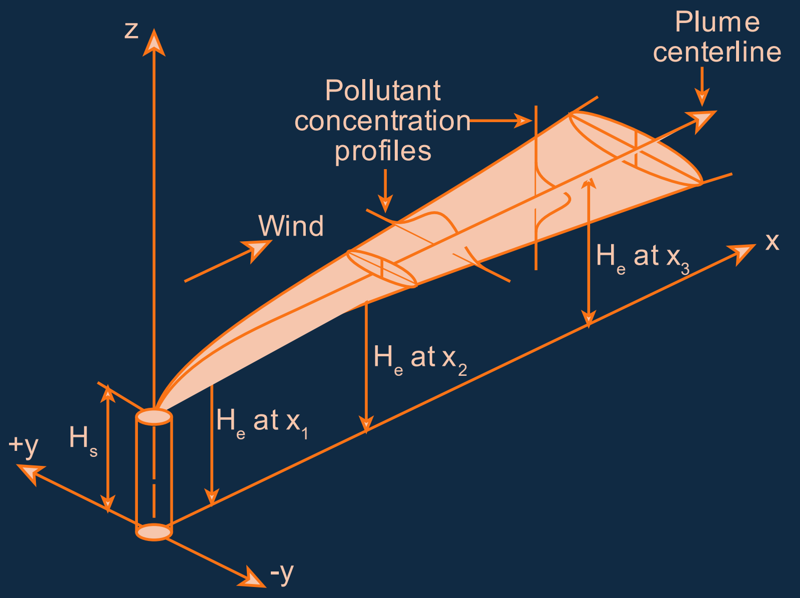
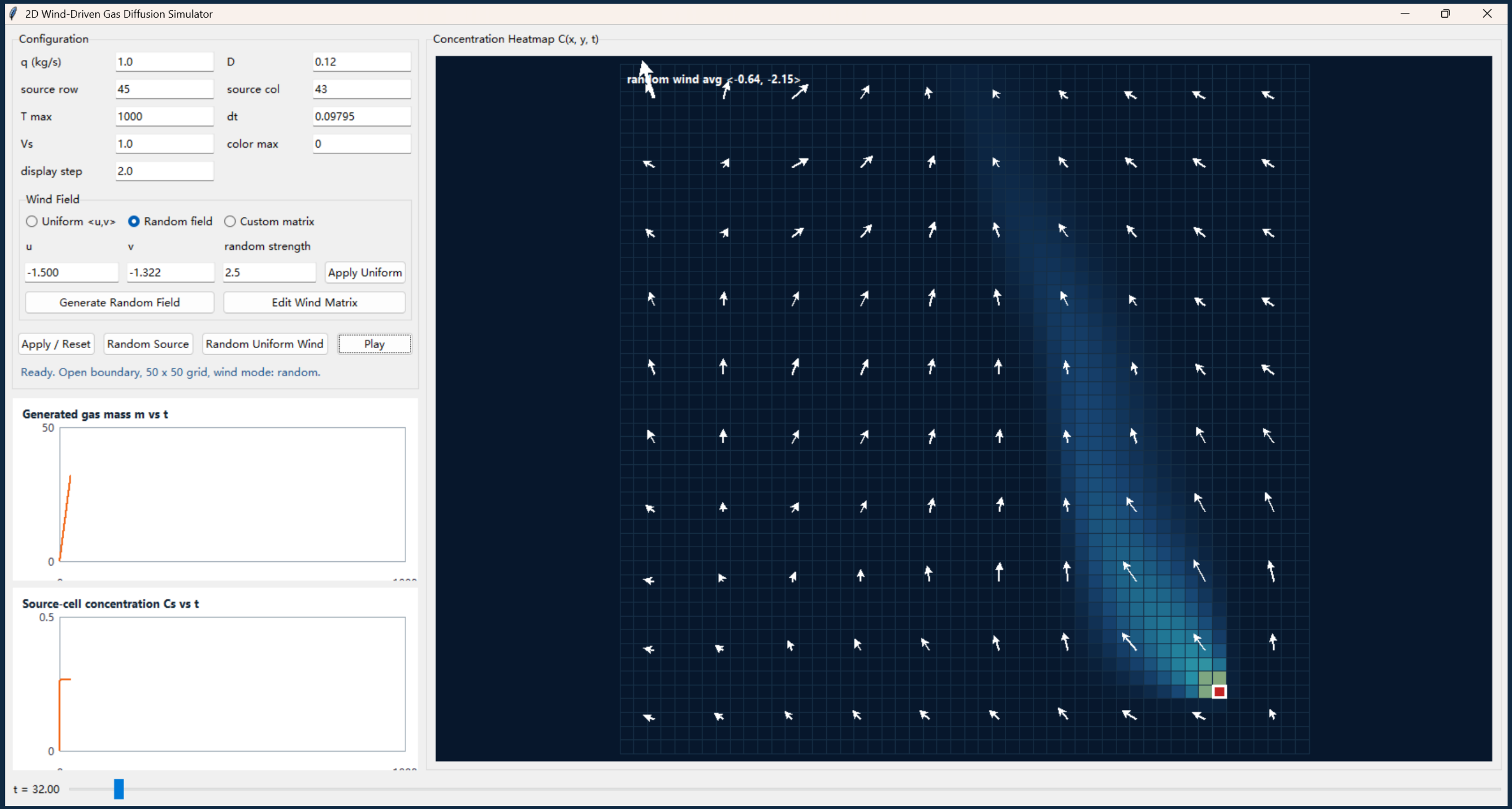


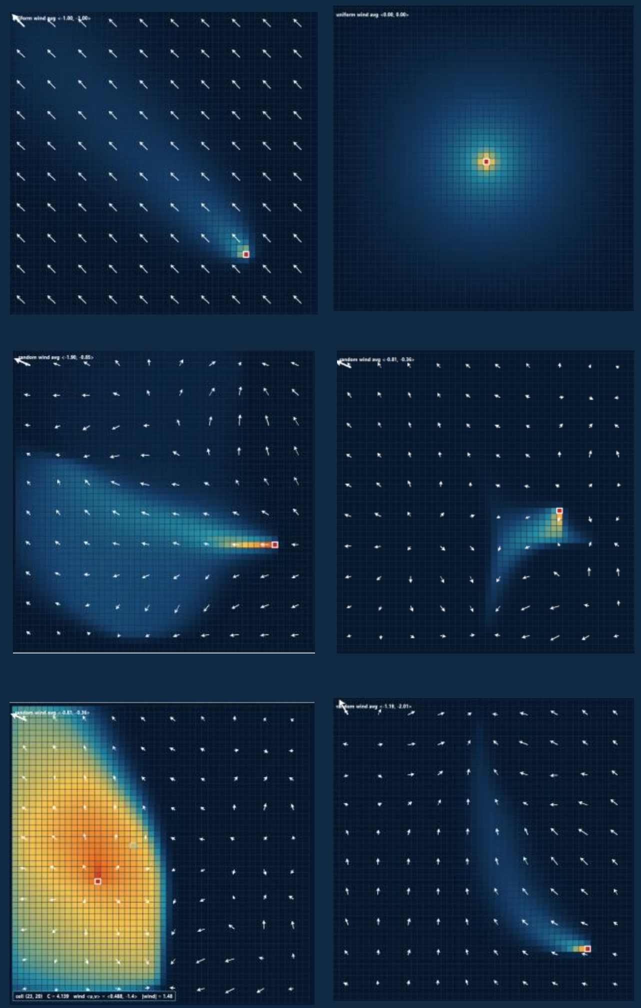
Fig 5. The Plume in 3D space (Daniel Case, 2018)

Step 2: 基于菲克定律的二维对流-扩散模型 模拟程序
Two-dimensional Convection-Diffusion Model Stimulator

北京一零一中
BEIJING 101 HIGH SCHOOL



(a) Main user's interface with input boxes, graphs and cells heat-map of concentration



(b)-(g) Cells heat-map of concentration in different conditions

Fig 6. Preview version of two-dimensional Convection-Diffusion Model Stimulator

Step 3: 目标函数与路径优化
Objective function and Path optimization

3a. 第一次建图 First Map Construction

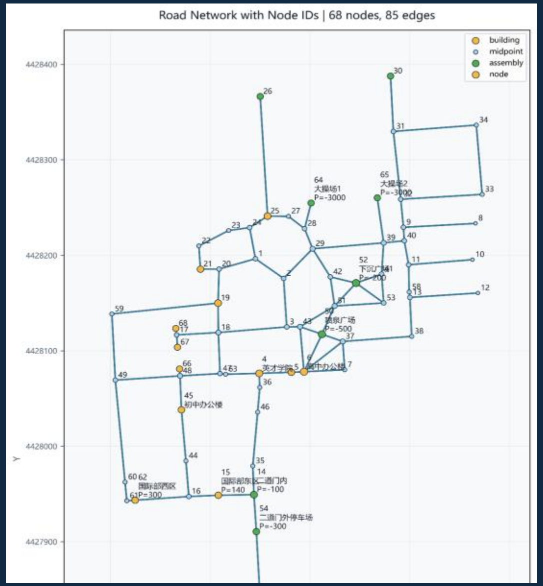
以***物理距离*** l **为边权建立静态边权的无向图**。图包括建筑节点、路径中间节点和疏散点；建筑节点和疏散点拥有人数属性P，所有建筑节点中的人需全部转移到疏散节点。

Construct an **undirected graph with *geographical distance*** l **as the static edge weight.** The diagram includes building nodes, intermediate nodes on the path, and evacuation points; building nodes and evacuation points have a population attribute P, and all people in the building nodes need to be evacuated to the evacuation nodes.

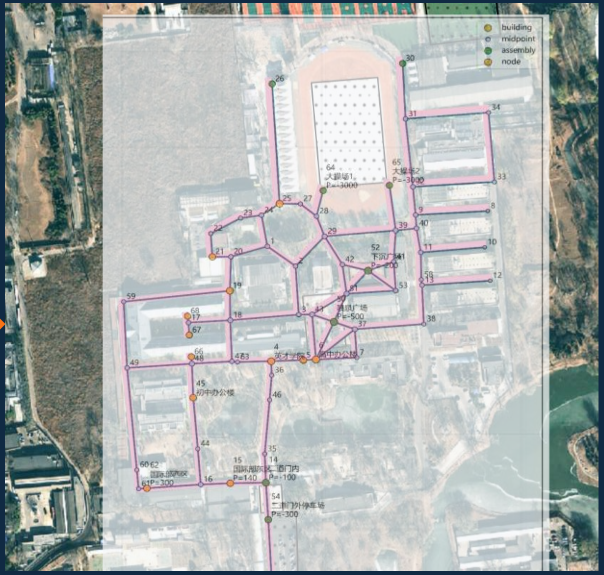
node_id	x	y	merged_count	label	P	type
1	440045.4	4428196	3			midpoint
2	440073.7	4428176	2			midpoint
3	440076.8	4428125	2			midpoint
4	440049.2	4428076	3	英才学院		building
5	440081.4	4428077	2			building
6	440094.1	4428078	5	高中办公楼		building
7	440135	4428080	2			midpoint
8	440266.1	4428233	1			midpoint
9	440193.9	4428229	2			midpoint
10	440263	4428195	1			midpoint
11	440199.1	4428190	2			midpoint
12	440268.5	4428160	1			midpoint
13	440200.3	4428156	2			midpoint
14	440043.9	4427949	3	二道门内	-100	assembly
15	440008.2	4427949	1	国际部东区	140	building
16	439978.6	4427947	3			midpoint
17	439966.4	4428117	3			midpoint
18	440008.2	4428119	4			midpoint
19	440007.9	4428150	3			building
20	440008.9	4428186	3			midpoint
21	439990.3	4428185	1			building
22	439988.7	4428210	1			midpoint
23	440018.6	4428226	1			midpoint
24	440039.6	4428229	3			midpoint
25	440057.6	4428241	3			building
26	440050.2	4428366	1			assembly
27	440078.5	4428241	1			midpoint
28	440094.7	4428228	2			midpoint
29	440102.8	4428207	4			midpoint
30	440180.9	4428388	1			assembly
31	440183.7	4428330	2			midpoint
32	440191.1	4428259	3			midpoint
33	440272.8	4428264	2			midpoint
34	440266.9	4428336	2			midpoint
35	440042.7	4427979	2			midpoint
36	440049.8	4428062	2			midpoint
37	440133	4428110	4			midpoint
38	440202.1	4428115	2			midpoint
39	440174	4428213	4			midpoint
40	440194.7	4428215	2			midpoint
41	440171.9	4428180	3			midpoint
42	440120.5	4428177	3			midpoint
43	440090.2	4428125	4			midpoint
44	439975.7	4427985	2			midpoint
45	439971.2	4428038	2	初中办公楼		building
46	440047.6	4428036	2			midpoint
47	440009.9	4428076	3			midpoint
48	439969.8	4428074	4			midpoint
49	439904.9	4428069	2			midpoint
50	440112.1	4428117	4	喷泉广场	-500	assembly
51	440124.8	4428147	5			midpoint
52	440146.3	4428171	4	下沉广场	-200	assembly
53	440174	4428150	3			midpoint
54	440046.3	4427911	1	二道门外停车	-300	assembly
55	440059.8	4427658	1	入校道	-1000	assembly
56	440094.6	4427602	1	一道门内	-100	assembly
57	440097.5	4427586	1	一道门外	-99999999	assembly
58	440199.6	4428162	2			midpoint
59	439901.4	4428139	1			midpoint
60	439914.7	4427962	1			midpoint
61	439916.5	4427943	1			midpoint
62	439924.8	4427943	1	国际部西区	300	building
63	440015.5	4428075	1			midpoint
64	440101.2	4428255	1	大操场1	-3000	assembly
65	440167.7	4428260	1	大操场2	-3000	assembly
66	439969.3	4428081	1			building
67	439967.2	4428104	1			building
68	439965.5	4428123	1			building

(a) Abstract points list

(b) The geographic map of Beijing 101 Middle School



(c) The undirected graph of Beijing 101 Middle School



(d) Pair edges and nodes with graph

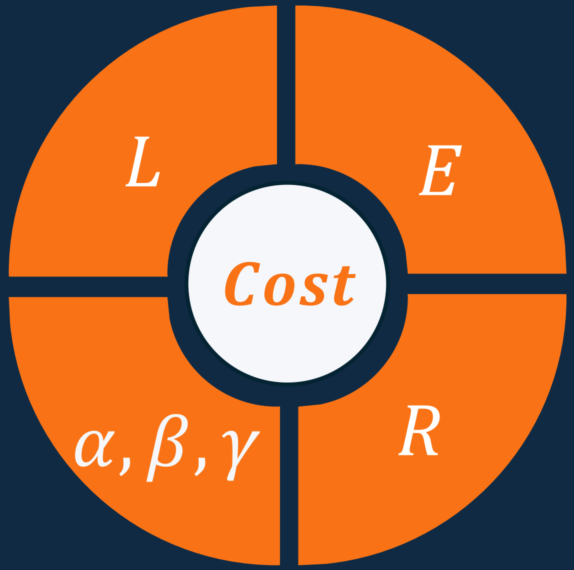
Fig 7. The processing of the first map construction

北京一零一中
BEIJING 101 HIGH SCHOOL

3b. 目标函数 Objective function

疏散风险来自疏散路径距离、暴露量、道路拥堵和集合点容量。为统一不同量纲的因素，研究将建立无量纲综合成本函数$Cost$，并通过路径长度L、暴露积分E和集合点风险R描述不同方案的差异。

The evacuation risk is influenced by factors such as evacuation route distance, exposure, road congestion, and assembly point capacity. To standardize factors with different dimensions, the study will establish a dimensionless comprehensive cost function $Cost$, and describe the differences between various scenarios through path length L, exposure integral E and assembly point risk R.



$$Cost = \alpha \cdot \frac{L}{L_0} + \beta \cdot \frac{E}{E_0} + \gamma \cdot \frac{R}{R_0},$$

$L = \sum l$,
$E = \int_0^T C(x(t), y(t), t)dt,$
$R = P_{will\ stay\ at\ final\ assembly\ point}$

Step 3: 目标函数与路径优化
Objective function and Path optimization

$$\min Cost = \alpha \cdot \frac{L}{L_0} + \beta \cdot \frac{E}{E_0} + \gamma \cdot \frac{R}{R_0},$$

$$L = \sum l\,,$$
$$E = \int_0^T C(x(t), y(t), t)dt,$$
$$R = P_{will\ stay\ at\ final\ point}$$

3c. 第二次建图 Second Map Construction

以$Cost$为边权建立随时间变化动态边权的无向图，将该问题转化为最短路径规划。

Construct an undirected graph with $Cost$ as the edge weight, representing dynamic edge weights that change over time, and transform the problem into shortest path planning.

3d. 路径优化 Path optimization*

可能使用*Multi-round Dijkstra*或*Time-Dependent A**算法求解。

Possible solutions include *Multi-rounds Dijkstra* or *Time-Dependent A* algorithm*.

Chapter 3.
Final Deliverables
最终递交物

最终递交物 Final Deliver Items

北京一零一中 BEIJING 101 HIGH SCHOOL

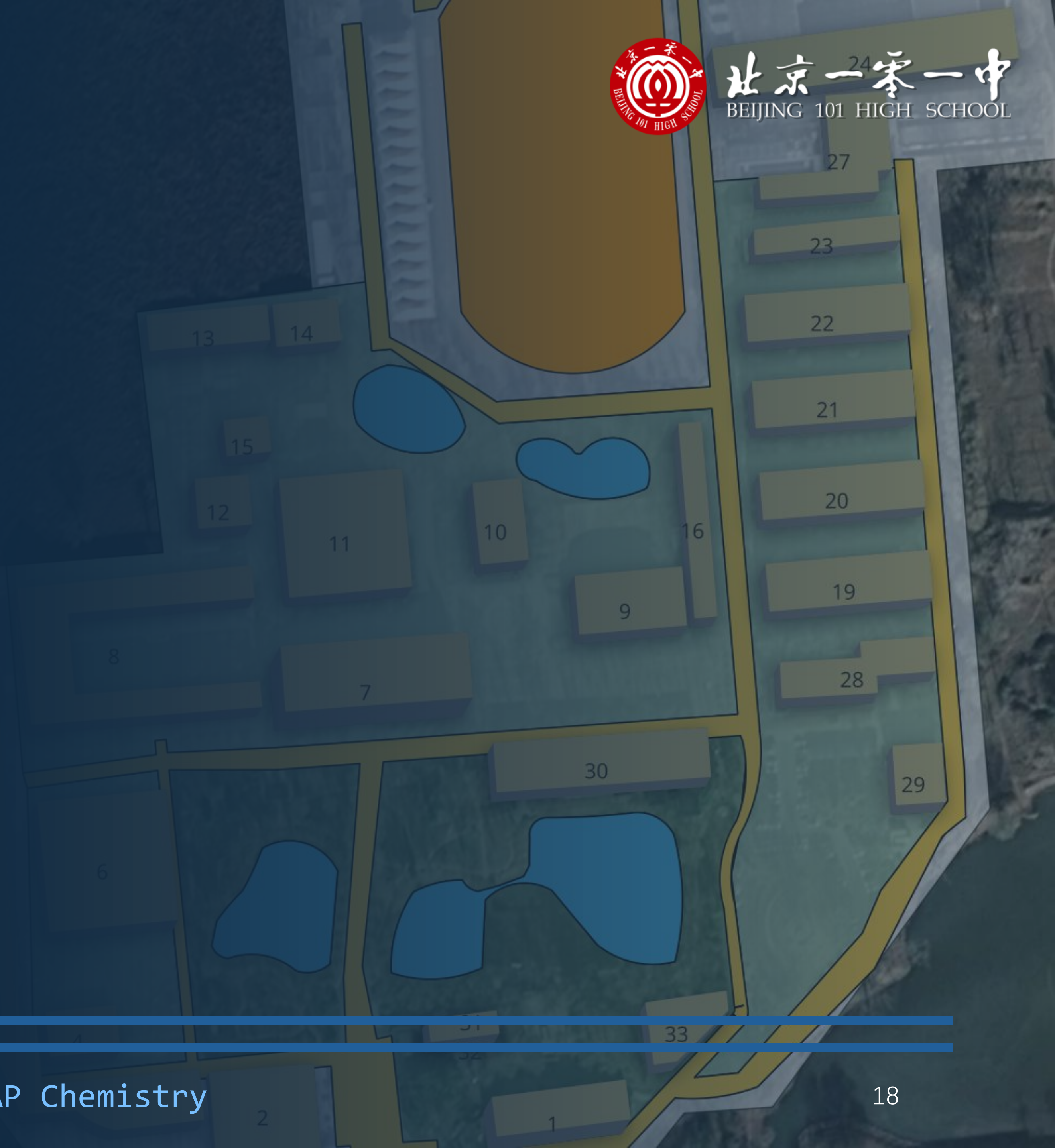
研究论文 Research Paper	海报 Poster	视频 Video	模拟程序 Simulation Program

本项目最终将以**四种形式**呈现研究成果：**论文、海报、视频和模拟程序**。论文将系统阐述研究背景、模型假设、化学分析、二维对流-扩散模型、路径优化目标函数以及最终方案评估；海报将以可视化方式总结研究问题、核心模型、流程图和预期结果；视频将用于动态展示火灾危险气体扩散、风险变化以及不同疏散路线和集合点选择的比较过程；模拟程序可用于进一步展示如何设计逃生路线。

The final outcomes of this project will be presented in **four formats: a research paper, a poster, a video, and a stimultion program.** The paper will systematically explain the research background, model assumptions, chemical analysis, two-dimensional advection-diffusion model, path optimization objective function, and evacuation plan evaluation. The poster will visually summarize the research question, core models, workflow, and expected results. The video will dynamically demonstrate hazardous gas diffusion, risk changes, and comparisons between different evacuation routes and assembly point choice; the simulation program can be used to further demonstrate how to design an escape route.

北京一零一中
BEIJING 101 HIGH SCHOOL

Chapter 4.
Project Feasibility
项目可行性



时间线 Timeline

最终提交时间 Final Deadline

本项目计划在**2026年6月第三周**完成最终递交。

The project is planned to be completed in the third week of June 2026.

第一阶段：资料整理与模型确定 | May Week 4

完成研究背景、化学原理、危险气体类型、二维对流-扩散模型和校园地图抽象方法的整理，明确研究假设与简化方案。

Collect background information, combustion chemistry, hazardous gas types, the 2D advection-diffusion model, and the simplified campus map representation.

第二阶段：模型计算与方案分析 | June Week 1

建立火源、风向、扩散范围、路径风险和集合点评价方式，比较不同情景下的疏散路线安全性。

Build fire source, wind direction, diffusion, path risk, and assembly point evaluation models, then compare evacuation safety under different scenarios.

第三阶段：论文、海报与视频制作 | June Week 2

完成论文初稿、海报主要版面和视频脚本，将化学模型、数学公式、地图图示和疏散方案整合为可展示成果。

Complete the paper draft, poster layout, and video script, integrating chemistry, mathematical formulas, campus maps, and evacuation plans.

第四阶段：修改、排版与最终提交 | June Week 3

检查逻辑完整性、双语表达、图表引用和格式规范，完成最终版本并提交论文、海报和视频。

Revise logic, bilingual wording, citations, figures, and formatting before submitting the final paper, poster, and video.

项目实用性分析

北京一零一中 BEIJING 101 HIGH SCHOOL

虽然学校发生大规模火灾的概率并不高，但火灾疏散方案不能只依赖发生概率低这一判断，而应体现**有备无患的安全管理思想**。提前研究不同火源位置、风向条件和危险气体扩散趋势下的疏散路径，有助于在事故真正发生前发现潜在风险，例如某些集合点可能位于下风向、部分道路可能经过烟气扩散区域，或常规疏散路线在特定情境下并不是最优选择。

本课题的实用价值在于提供一种**可推广的分析方法。**通过将人员密集场所建筑抽象为空间节点，将道路抽象为连接边，再结合危险气体释放与二维扩散模型，可以比较不同疏散路线和集合点在多种情景下的安全性。这种方法能够帮助人员密集场所管理从经验设计转向理论分析，使疏散方案更加透明、可解释，也更容易根据新建筑、新道路或新安全要求进行更新。

此外，该方法**并不限于学校场景**。凡是由多个建筑、道路和集合区域组成的空间系统，例如大学校园、住宅小区、商业园区、医院、办公园区，甚至化工厂和仓储区，都可以采用类似思路进行疏散方案分析。尤其在工厂火灾或含有特殊材料、化学品的建筑群中，危险气体的释放与风向影响可能更加显著，因此基于风驱动气体扩散模型的路径与集合点评估具有更强的现实意义。

换言之，本课题以校园为例，但研究目标是建立一种面向建筑群火灾疏散的通用设计方法。

Project practicality analysis

Although the probability of a large-scale fire occurring in a school is low, the fire evacuation plan should not solely rely on this low probability judgment, but rather embody the safety management philosophy of being prepared for any eventuality. Studying evacuation routes under different fire source locations, wind direction conditions, and hazardous gas diffusion trends in advance can help identify potential risks before an accident actually occurs. For example, some assembly points may be located in the downwind direction, some roads may pass through smoke diffusion areas, or conventional evacuation routes may not be the optimal choice under specific circumstances.

The practical value of this research lies in providing a generalizable analytical method. By abstracting buildings with dense personnel as spatial nodes and roads as connecting edges, and combining this with a model for hazardous gas release and two-dimensional diffusion, it is possible to compare the safety of different evacuation routes and assembly points under various scenarios. This method can assist in transitioning the management of densely populated areas from empirical design to theoretical analysis, making evacuation plans more transparent, interpretable, and easier to update in response to new buildings, roads, or safety requirements.

Furthermore, this method is not limited to school settings. Any spatial system composed of multiple buildings, roads, and gathering areas, such as university campuses, residential areas, commercial parks, hospitals, office parks, and even chemical plants and storage areas, can adopt a similar approach for evacuation plan analysis. Especially in factory fires or building complexes containing special materials or chemicals, the release of hazardous gases and the influence of wind direction may be more significant. Therefore, the evaluation of paths and assembly points based on wind-driven gas diffusion models has stronger practical significance.

In other words, this project takes the campus as an example, but the research objective is to establish a general design method for fire evacuation in building complexes.

备用方案 Backup Plan

核心原则 Core Principle

即使部分模型被简化，项目仍保留完整主线：
燃烧化学 → 危险气体释放 → 风驱动二维扩散 → 风险成本 → 疏散路线与集合点建议。

Even if some modules are simplified, the project will preserve the full research chain: combustion chemistry → hazardous gas release → wind-driven 2D diffusion → risk cost → route and assembly point recommendation.

简化方法 Simplified Methods

1. 参数简化 Parameter simplification;
2. 若真实风场难以获取，扩散模型将先使用恒定风向量 If the actual wind field data is difficult to obtain, the diffusion model will initially utilize a constant wind vector;
3. 若动态寻路算法无法完全实现，可改为比较$3-5$条候选路线，并选择目标函数$Cost$最小的路线，而不是实时计算全局最优路径 If the dynamic pathfinding algorithm cannot be fully implemented, one can compare $3-5$ candidate routes and select the route with the minimum objective function $Cost$, rather than calculating the globally optimal path in real-time;
4. 最小可交付成果：一套简化公式、一个二维扩散热力图、一个校园道路图模型、一套基于风险成本的疏散路线/集合点建议，以及海报 Minimum deliverables: a set of simplified formulas, a two-dimensional heat diffusion diagram, a campus road map model, a set of evacuation route/assembly point suggestions based on risk cost, and posters.

北京一零一中
BEIJING 101 HIGH SCHOOL

Chapter 5.
References
参考列表

References List

Arumugam, G. P., Augustine, J., Golin, M. J., Higashikawa, Y., Katoh, N., & Srikanthan, P. (2021). *Optimal evacuation flows on dynamic paths with general edge capacities*. arXiv. https://arxiv.org/abs/1606.07208

Gann, R. G., Averill, J. D., Johnsson, E. L., Nyden, M. R., & Peacock, R. D. (2010). *Fire effluent component yields from room-scale fire tests*. Fire and Materials, 34(6), 285-314. https://doi.org/10.1002/fam.1024

Hart, P. E., Nilsson, N. J., & Raphael, B. (1968*). A formal basis for the heuristic determination of minimum cost paths*. IEEE Transactions on Systems Science and Cybernetics, 4(2), 100-107. https://doi.org/10.1109/TSSC.1968.300136

Kemloh Wagoum, A. U., Seyfried, A., & Holl, S. (2012). *Modeling the dynamic route choice of pedestrians to assess the criticality of building evacuation*. Advances in Complex Systems, 15(7), Article 1250029. https://doi.org/10.1142/S0219525912500294

Koenig, S., & Likhachev, M. (2002). *D* Lite. In Proceedings of the AAAI Conference on Artificial Intelligence* (pp. 476-483). AAAI Press. https://cdn.aaai.org/AAAI/2002/AAAI02-072.pdf

McGrattan, K., Hostikka, S., McDermott, R., Floyd, J., Weinschenk, C., & Overholt, K. (2013). *Fire Dynamics Simulator technical reference guide, Volume 1: Mathematical model* (NIST Special Publication 1018, 6th ed.). National Institute of Standards and Technology. https://doi.org/10.6028/NIST.SP.1018e6

McKenna, S. T., & Hull, T. R. (2016). *The fire toxicity of polyurethane foams*. Fire Science Reviews, 5, Article 3. https://doi.org/10.1186/s40038-016-0012-3

Stockie, J. M. (2011). *The mathematics of atmospheric dispersion modeling*. SIAM Review, 53(2), 349-372. https://doi.org/10.1137/10080991X

Thanks!

开题报告
Research Proposal

Authors:
陈禹翔 Lambert Yuxiang Chen[1]
彭博 Blake Bo Peng[1]
[1]Beijing 101 Middle School International Department